ADSI – AI Drug Stability Intelligence System

Predictive Monitoring of Pharmaceutical Degradation Using AI and Environmental Data

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Abstract

Pharmaceutical drug stability is a critical factor in ensuring treatment effectiveness and patient safety. In many developing regions, particularly in tropical climates such as Nigeria, drugs are exposed to extreme environmental conditions including high temperature and humidity during storage and transportation. These conditions accelerate chemical degradation, reducing potency and increasing health risks.

This project presents ADSI (AI Drug Stability Intelligence System), an AI-powered monitoring system that predicts drug degradation risk based on environmental exposure data. ADSI integrates machine learning, IoT-based sensing, and a real-time dashboard to provide actionable insights for pharmacies, regulators, and supply chain stakeholders.

The system supports both manual inspection input and automated sensor-based monitoring, enabling flexible deployment across environments with varying technological infrastructure. A unified risk assessment engine generates compliance status, alerts, and recommended actions to prevent distribution of degraded drugs.

1. Introduction

Drug stability directly affects the safety and effectiveness of medications. In many African countries, pharmaceutical supply chains lack proper environmental monitoring systems, leading to exposure of drugs to harmful conditions.

Traditional approaches rely on:

- Laboratory testing (after damage has occurred)
- Manual inspection without predictive capability

ADSI introduces a proactive approach by predicting degradation risk before drugs become unsafe.

2. Problem Statement

Pharmaceutical degradation occurs due to:

- High temperature exposure
- High humidity
- Extended storage duration

Challenges include:

- Lack of real-time monitoring
- Poor infrastructure in rural and urban areas
- Absence of predictive systems
- Limited regulatory visibility

3. Proposed Solution

ADSI provides:

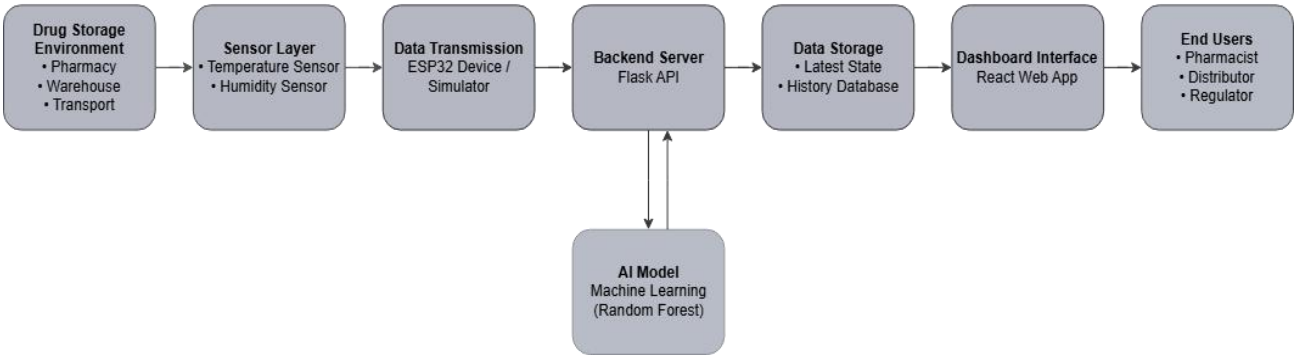
- AI-based degradation prediction
- Real-time environmental monitoring
- Risk classification (Low, Moderate, High)
- Compliance tracking
- Alert generation and escalation

4. System Architecture

ADSI follows a modular architecture:

Sensor / Manual Input → Backend API → AI Model → Risk Engine → Dashboard

System Architecture Diagram



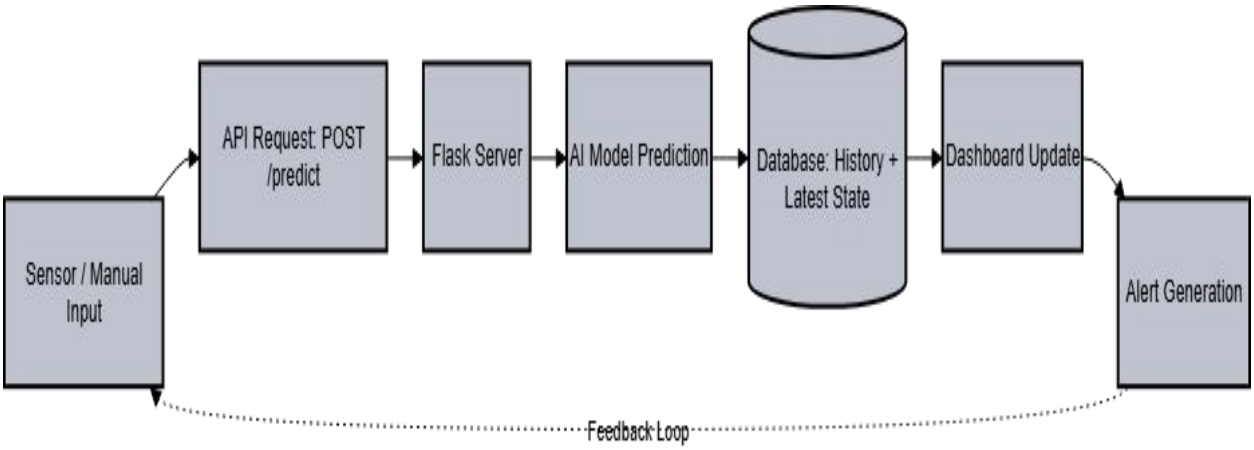
5. Data Design

The system uses a structured dataset:

- drug_name
- drug_class
- temperature_exposure
- humidity
- exposure_hours
- stability_index
- risk_label

Synthetic data is used for training to simulate real-world environmental exposure.

Data Flow / Pipeline Diagram



6. AI Model

The model predicts degradation risk using:

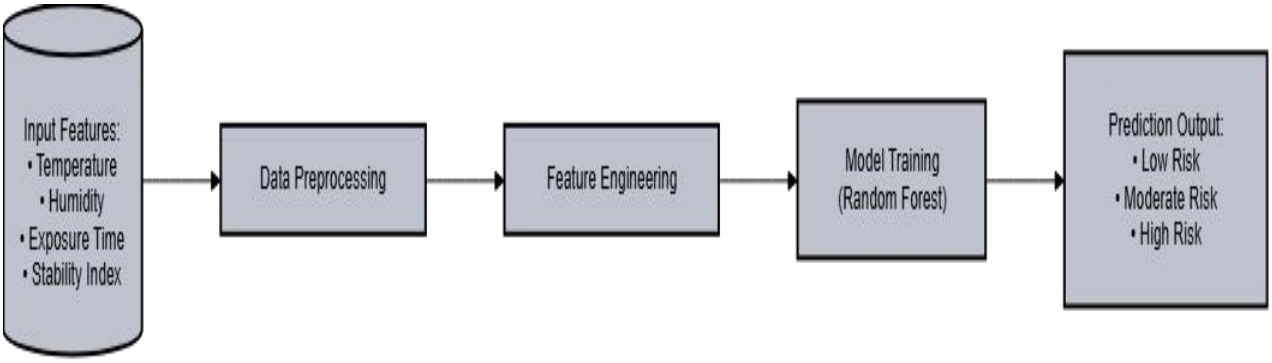
Input: environmental conditions

Output: risk classification

Approach:

- Supervised learning (scikit-learn)
- Classification model

AI Pipeline Diagram



7. Model Validation and Domain Justification

7.1 Model Approach

The ADSI model predicts drug degradation risk based on environmental inputs including temperature, humidity, and exposure time. A supervised learning approach was used to classify risk levels into Low, Moderate, and High categories.

7.2 Domain Justification

Pharmaceutical stability is highly dependent on environmental conditions. According to WHO guidelines, exposure to high temperature and humidity accelerates chemical degradation of drugs, reducing their effectiveness.

Similarly, NAFDAC emphasizes proper storage conditions as a critical requirement for maintaining drug quality throughout the supply chain.

These principles guided the design of the dataset and model features used in ADSI.

7.3 Model Validation

The model was evaluated using standard classification metrics to assess its performance.

- Accuracy: 98.4%
- Precision: 98.0%
- Recall: 98.0%

These results indicate that the model can reliably classify degradation risk under different environmental conditions.

7.4 Limitations

The current model is trained on a structured synthetic dataset due to limited availability of real-world pharmaceutical stability data.

7.5 Future Improvements

Future work includes integrating real IoT sensor data and laboratory-validated measurements to further improve prediction accuracy and reliability.

8. System Components

8.1 Manual Input Module

Used for:

- inspections
- non-sensor facilities

Supports:

- facility ID
- source ID
- batch ID

8.2 *Sensor Monitoring Module*

- ESP32
- DHT22 sensors

It provides continuous environmental data

In this prototype, sensor data is simulated but follows real IoT structure.

8.3 Unified Risk Assessment Engine

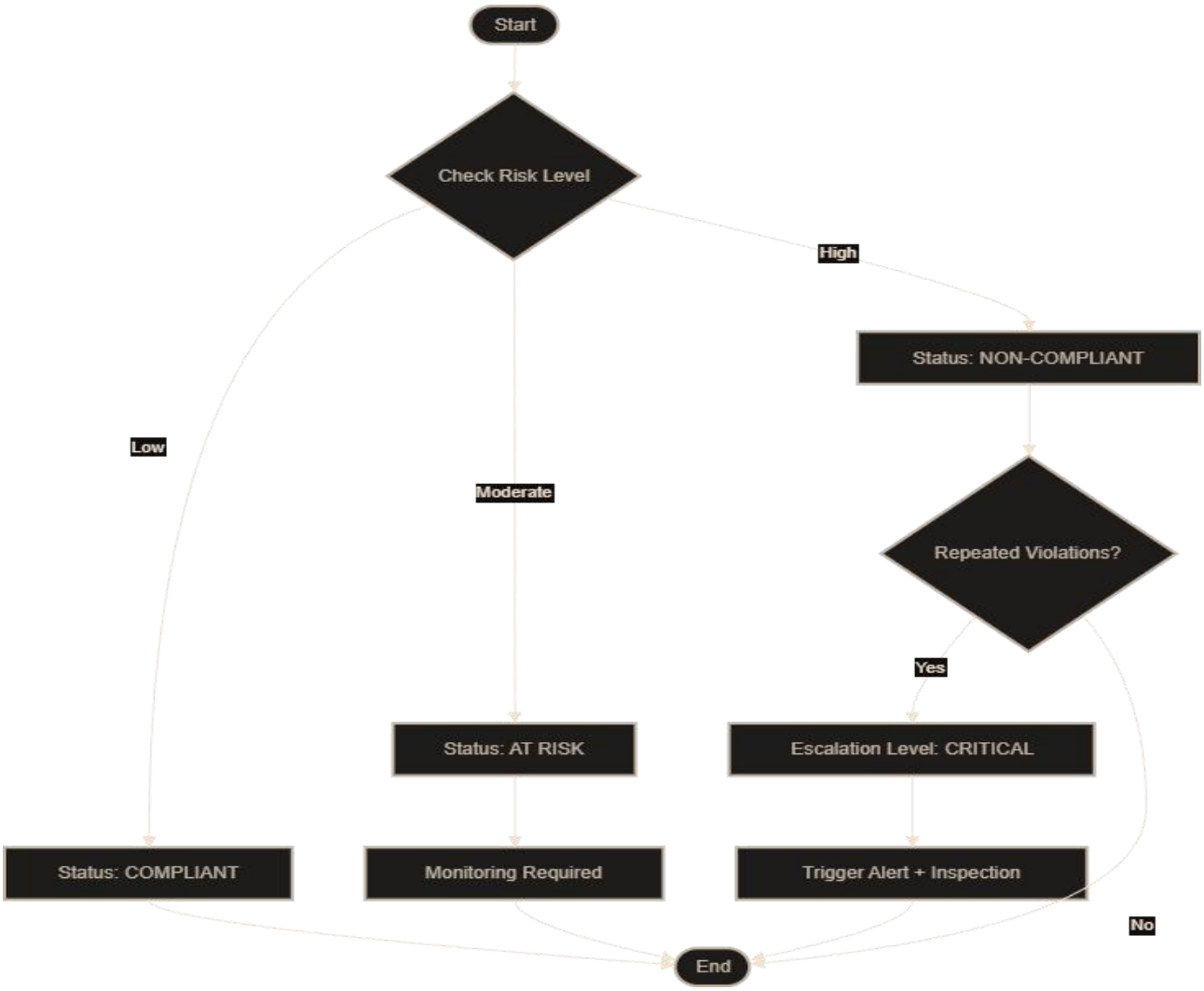
Combines:

- manual input
- sensor data

Outputs:

- risk level
- compliance status
- breach detection
- recommended actions

Escalation Logic / Risk Engine Diagram

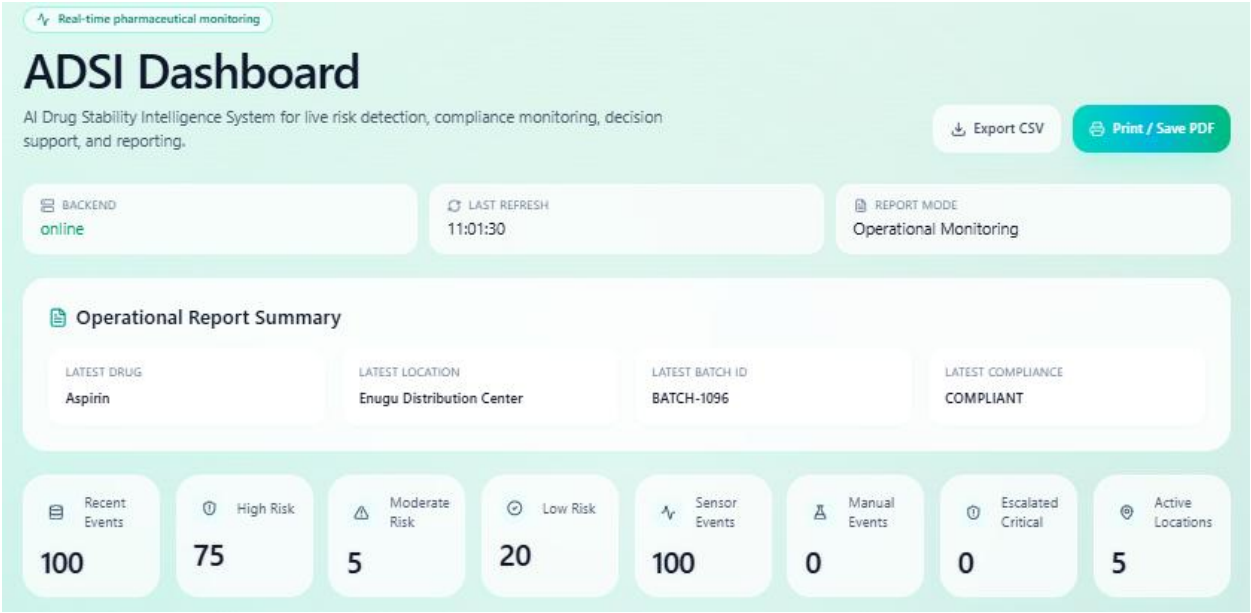


8.4 Dashboard Interface

Features:

- manual input panel
- sensor data panel
- unified monitoring assessment
- alert center
- trend charts
- inspection priority system

Dashboard Overview

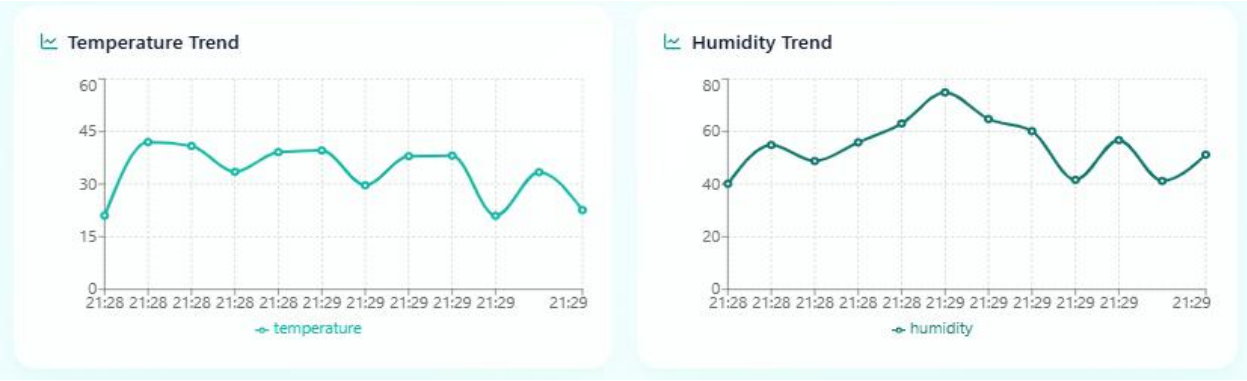


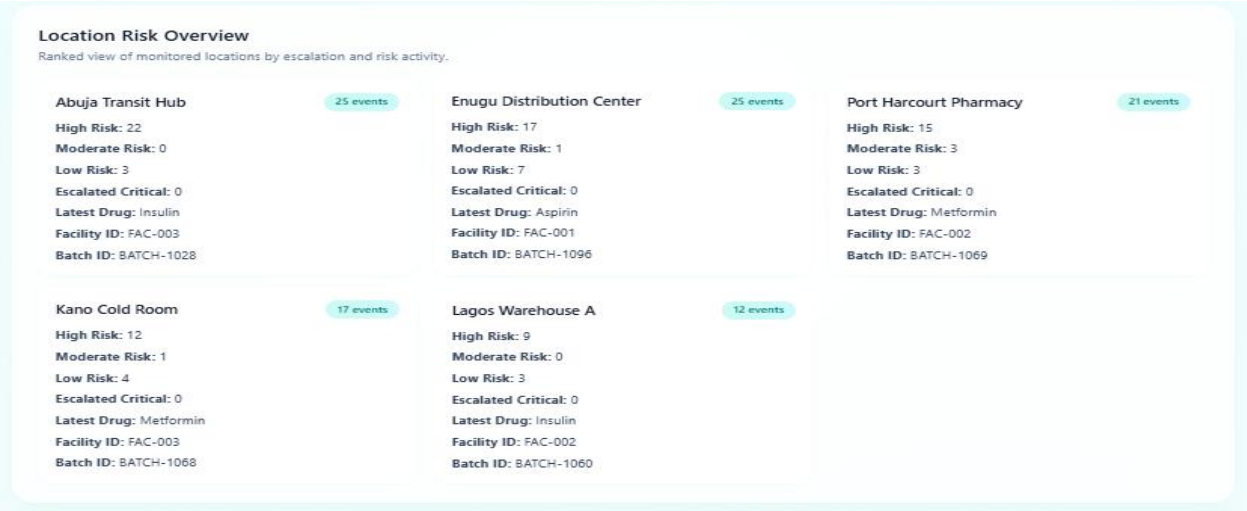
9. Results and Features

The system provides:

- real-time monitoring
- predictive risk scoring
- automated alerts
- location-based risk analysis
- inspection prioritization

Charts (temperature, humidity, risk distribution)





10. Deployment

Frontend (UI)

- React (Vite)
- Tailwind CSS

Backend (App Engine):

- Flask (Python)

Deployment:

- Vercel (frontend)
- Render (backend)

Simulation:

- local Python simulator for live data stream that mimic the real world IOTs.

11. Benchmarking and Uniqueness

Compared to traditional systems:

Feature	**Traditional**	**ADSI**
Monitoring	Manual	Real-time
Prediction	None	AI-based
Alerts	Reactive	Proactive
Coverage	Limited	Scalable

Uniqueness:

- Combines AI + IoT + dashboard
- Works with or without sensors
- Designed for African climate realities

12. Impact

ADSI enables:

- early detection of drug degradation
- improved patient safety
- reduced financial losses
- better regulatory oversight

12. Conclusion

ADSI demonstrates how AI and environmental monitoring can transform pharmaceutical safety systems. By providing predictive intelligence and real-time insights, the system supports safer drug distribution and improved healthcare outcomes.